Assignment 2 - 1: Organizing the Research Data

1. Introduction & Scenario

Dr. Aris Thorne, the lead researcher for Project Starlight, is impressed with your work in setting up the initial S3 bucket. She has just given you a new set of data from a different telescope array: raw sensor readings, her personal observation notes, and a newly processed image of the Orion Nebula.

Your assignment is to create a *new*, separate S3 bucket for this dataset and upload the files to a specific folder structure. You must complete this entire task using only the commands you learned in **Lab 2**, operating within AWS CloudShell.

2. Learning Objectives

This assignment will assess your ability to:

- Independently use basic S3 CLI commands (`mb`, `cp`, `ls`).
- Translate a set of requirements into the correct sequence of commands.
- Create a specific folder (prefix) structure within an S3 bucket.
- Verify your work using CLI commands.

3. Your Task & Requirements

You must perform the following steps in order within AWS CloudShell.

Step 1: Prepare the Data in CloudShell

First, you need to create the new data files inside your CloudShell environment. Copy and paste these commands into your terminal to create the files.

1. **Create a new directory and move into it:**

   ```
   mkdir research_data && cd research_data
   ```

2. **Create the sensor data file:**

   ```
   echo
   "Timestamp,SensorID,Reading\n2025-09-17T01:00:00Z,Alpha,4.51\n2025-09-1
   7T01:05:00Z,Beta,10.2" > sensor-feed.csv
   ```

3. **Create the observation notes file:**

```
echo "Observation Notes for Sept 17, 2025. Atmospheric conditions were clear.
The Orion Nebula was exceptionally visible tonight." > 2025-09-17-notes.txt
```

4. **Download the new nebula image:**

   ```
   wget https://esahubble.org/media/archives/images/large/heic1907a.jpg -O
   orion-nebula.jpg
   ```

Step 2: Create and Organize the S3 Bucket

Now that your files are ready, your main task is to create an S3 bucket and upload them to the correct locations.

1. **Create a new S3 bucket.** The bucket name must be globally unique and follow this format: `starlight-research-yourname-` followed by the current date (e.g., `starlight-research-janedoe-20250917`).
2. **Upload the files to the bucket.** You must use the `aws s3 cp` command to place each file into a specific "folder" (prefix). When you are finished, your bucket must have the following exact structure:
    - `/raw-data/sensor-feed.csv`
    - `/observation-notes/2025-09-17-notes.txt`
    - `/processed-images/orion-nebula.jpg`

Step 3: Verify Your Work

To confirm that you have organized the bucket correctly, run the following command, replacing `your-bucket-name` with the actual name of the bucket you created.

```
aws s3 ls s3://your-bucket-name --recursive
```

The output of this command should clearly show the three files organized under their correct prefixes.

4. Deliverables

Submit a document containing the following two items:

1. **Your S3 Bucket Name:** The full, unique name of the `starlight-research-...` bucket you created.
2. **Verification Screenshot:** A screenshot of your CloudShell window showing the output of the final verification command from Step 3.

5. Academic Integrity

This is an individual assignment. You are expected to use the commands and concepts from the previous labs to complete the required tasks. Your submission must be your own work.